Raghavan 2021 PDAC — analysis summary

RUN 4 OF 5

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony θ-sweep resolves a per-patient archipelago (θ=0, iLISI 1) into one continuous well-mixed mass (θ=3, iLISI 4.8/21). Island-free (largest component 99.1%; 72-cell speck removed).

Gate 2 — module geography → GO (non-branching)

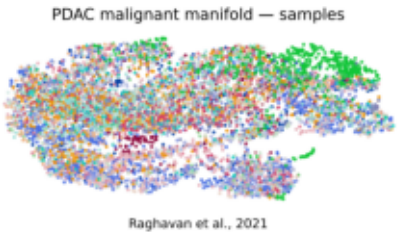
Single embryonic axis: Gast/pSom co-vary at the origin (r=+0.12), both oppose Crest toward the terminus (Gast↔Crest −0.50, pSom↔Crest −0.61; pSom&Crest joint-high enrichment 0.16). No two-arm hub.

Gate 3 — hub cross-repression → NA

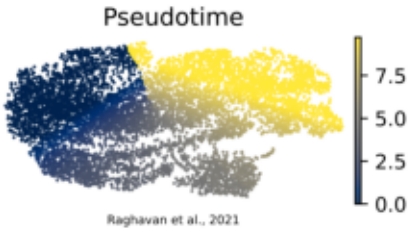
Curve has 0 forks (PPT tree forks unstable 2→3 across N — round-blob overfit). No branch to test.

STEP FIGURES

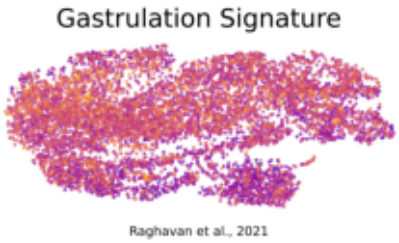
Manifold — samples



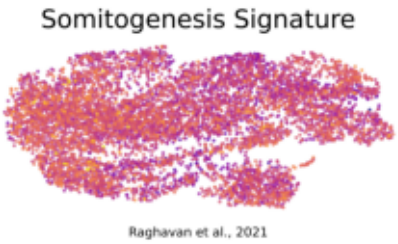
Pseudotime



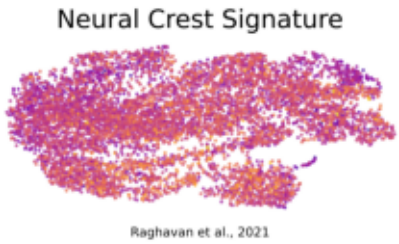
Gastrulation



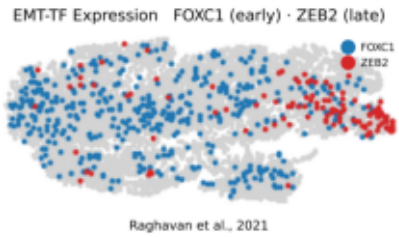
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Metastatic PDAC (Raghavan et al., 2021), Seq-Well — 7,904 malignant cells → 7,830 after 100/sample threshold (21/23 samples) → 7,758 after island removal. Subtypes: Classical 3,314 · Mixed 3,341 · Basal 1,103.

Verdict.
The embryonic pattern reproduces as the non-branching case — one continuous Gast/pSom-origin → Crest-terminus axis with pSom/Crest mutual exclusivity; developmental EMT-TF order FOXC1 → ZEB2. No bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means Gast −0.011 / pSom +0.072 / Crest −0.024.

Parameter choices

Min cells/sample	100 (gap 45→120; 21/23 kept)	**Integration**	Harmony θ=3 · n_pcs=30 · spread=1.3
Gene selection	1,497 overdispersed (find_overdispersed)	**Scope**	whole manifold (cycling |ρ|≤0.19, orthog.)
Trajectory	scFates curve Nodes=20, λ=0.3, μ=100; fold-free (SA 2.56); 0 forks	**Root**	tip 0 = Gast/pSom-high, Crest-low (dev. origin)
Subtype source	Raghavan 2021 · Moffitt basal/classical	**EMT-TFs**	FOXC1 (early) · ZEB2 (late)